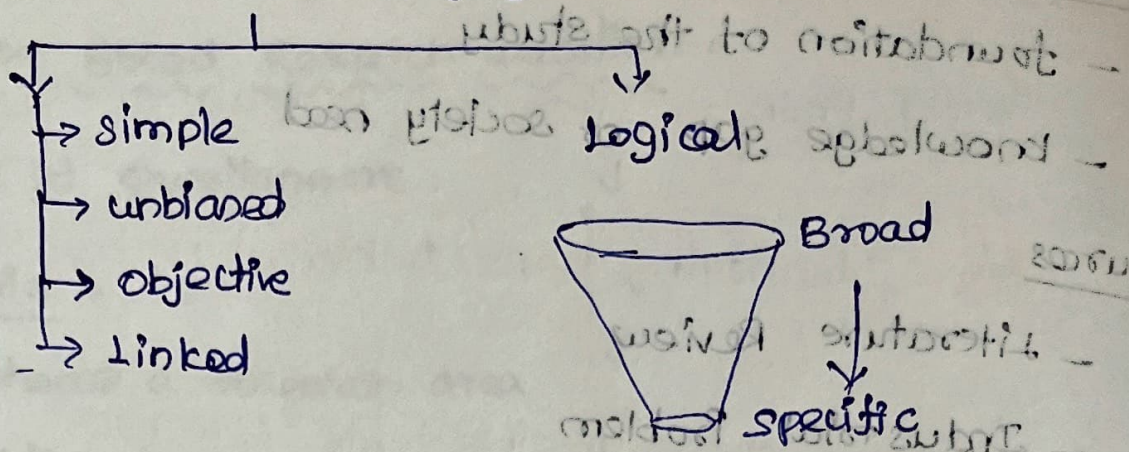


Questions category



— Pilot Testing : small no. of samples are testing —

Characteristics of Good Research

- clearly explain
- systematic Process
- Empirical
- Replicability
- objective and neutrality
- Ethical practices
- usefulness and applicability
- originality and innovative.

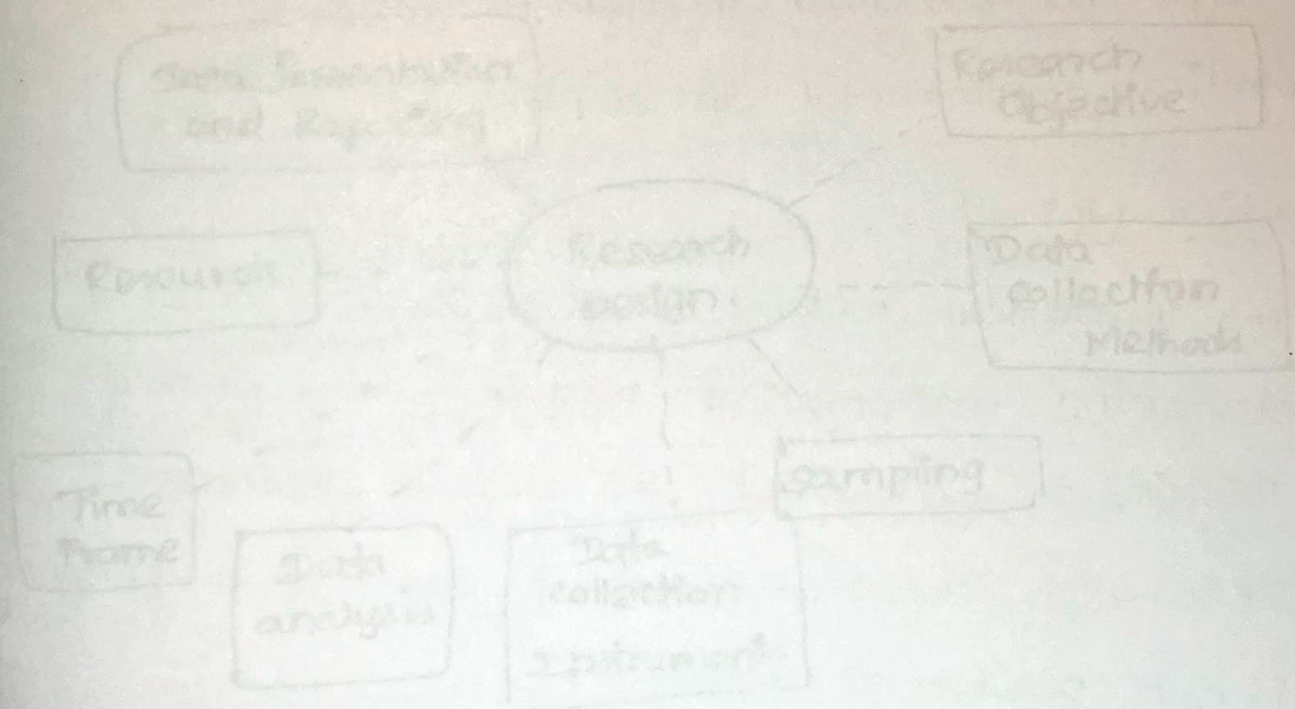
08.09.25

characteristics of Good Research

A Good research study is one that is systematic, logical and purposeful and meets academic or scientific

It lays out the procedures for collecting, measuring and analyzing data in a way that answers the research question or tests the hypothesis.

A good research design ensures that the evidence obtained enables the researcher to effectively address the research problem with accuracy and efficiency.

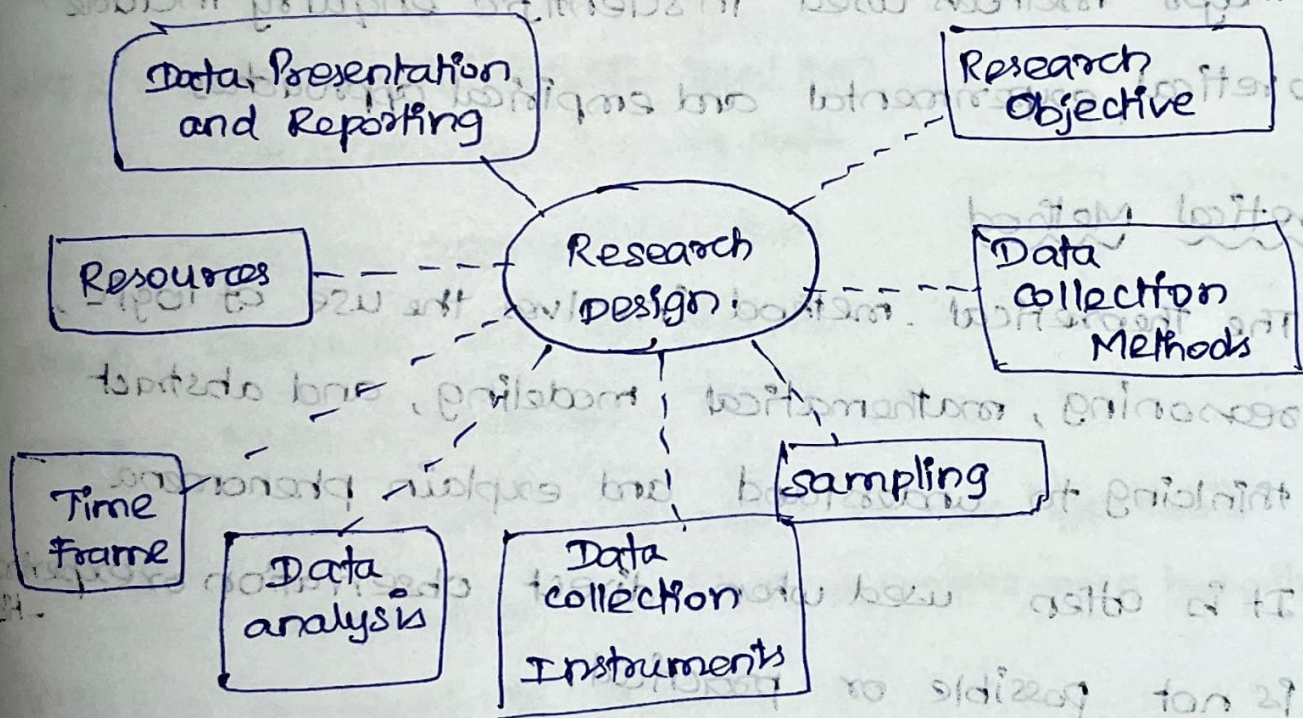


Research design

Research design is the framework or blueprint for conducting a research project.

It lays out the procedures for collecting, measuring and analyzing data in a way that answers the research questions or tests the hypotheses.

A good research design ensures that the evidence obtained enables the researcher to effectively address the research problem with accuracy and efficiency.



Methods of Scientific Enquiry:

* Scientific enquiry refers to the systematic investigation of phenomenon through object observation, experimentation and logical reasoning to discover new knowledge or validate existing one

* It is the foundation of research in natural and social sciences and it follows a structured and objective approach to understand how things work.

* The major methods used in scientific enquiry include theoretical, experimental and empirical approaches.

1. Theoretical Method

→ The theoretical method involves the use of logic, reasoning, mathematical modeling, and abstract thinking to understand and explain phenomena.

→ It is often used when direct observation or experimentation is not possible or practical.

→ Theoretical enquiry helps in developing models, hypotheses and predictions that can later be tested through empirical method.

2. Experimental Method :

Example problem.

A researcher wants to study whether a new Mg-based alloy reduces hydrogen absorption time compared to the existing alloy.

Population mean absorption time: $\mu = 4200 \text{ sec}$

Testing of Hypothesis :

Step 1 : Formulate the hypothesis

Step 2 : Select a significant level (α)

α is always 5% of data

Step 3 : choose the appropriate test

Step 4 : Compute the test statistic

Step 5 : Make a decision.

If $p\text{-value} \leq \alpha$, reject the null hypothesis

If $p\text{-value} > \alpha$, fail to reject null hypothesis.

Step 6 : Interpret the results

Sources of Information - ICT Enabled Tools.

- Revolutionized how researchers access, manage and disseminate information.

1) Digital Libraries : Access academic papers and books
Eg : IEEE Xplore, Shodhganga,

2) Online Journals : Latest research articles and reviews
Eg : ScienceDirect, SpringerLink etc.

3) Search Engines : Find scholarly content quickly
Eg : Google Scholar, Scopus

4) Reference Managers - organize and cite sources

Eg : Mendeley, Zotero

5) Online surveys : Collect data via forms

Eg : Google Forms, SurveyMonkey

6) E-learning Platforms : Learn research methods and tools

Eg : NPTEL, Coursera

7) Academic Networks : Connect with researchers

Eg : ResearchGate, Academia.edu.

1) Database : scopus , was

2) Repositories : so shodhganga , Dryad

3) Public sources : WHO ,

4) Private sources : Elsevier.

15.09.2025.

L&T = 15 marks.

AT1 = 10 + 5 = 15 marks = OCT 6th 2025.

1) Databases :

scopus , was , Pubmed, → can search the documents and scientific data.

2) Repositories

Digital archives for storing research outputs like theses , reports , data.

Eg : Shodhganga (India) , arXiv , Dryad.

3) Public sources

Freely available government or institutional data and reports

Eg : Census data , National sample survey , WHO , world Bank

4) Private sources :

Commercial or subscription based information services

Eg : Gartner reports , Elsevier , Bloomberg , Market research firms.

Indexes.

- systematically organize and locate published literature
- Index typically refers to bibliographic databases that allow users to search for articles, papers, journals and books based on authors, titles, subjects or keywords.
Eg: scopus, wos, Pubmed

Literature search - keywords

- A keyword-based literature search is one of the most fundamental methods to identify relevant research papers, articles or books.
- It involves selecting appropriate keywords or key phrases that represent the core idea of research

Backward and forward search

1) Backward search :

Also referred as reference mining, this involves going through the reference list of a relevant paper to discover earlier work that it is cited.

This helps in understanding the background and theoretical framework that led to the current study.

Forward Search :

This method tracks who has cited a particular paper after its publication. It helps in identifying newer studies that have built upon the original work.

statement - I

A startup wants to understand what features EV buyers

value most. - Fwd search.

statement - II.

A car maker^{is} deciding whether to add solar charging

panels to its next EV models. - Bwd search.

Difference b/w Fwd & Bwd search

Aspect	Fwd Research	Bwd Research
Definition	starts with curiosity or a research qn and works forward through exploration to discover	desired outcome or decision and works backward to determine
Focus	New understand	specific problem

Quality Measurement in Research

Evaluating the quality and impact of research is

crucial to ensure its validity, relevance and contribution to the academic community.

Journal Metrics

$$IF = \frac{\text{Avg. No. papers published in past 2 years}}{\text{total No. of papers cited - items published in past 2 yrs}}$$

* Artificial Influential score.

* Print ISSN. - hard copy & soft copy both.

* e-ISSN - soft copy only.

* Quartile. - Journal rating.

* H index

* publishers

* Areas covering

* SJR

* SNIP.

* country.

FILED IN 2734

→ Higher citation rates

have at least 11 channels

carbons with a
sum of 10

(9142)

200,000 Normalized

on scope to estimate impact of a journal

From impact factor (151) 202

with strong oscillation

Abstract of Elias collection / page 107

9142
In 2022, 2023,

1. 1000000

in 2023 = 170

total $M = 300$ bni oji

in 2022 & 2023

600 $\Rightarrow 2 \rightarrow$ of 2024.

$$\frac{\text{Total no. of citation fetched by a journal}}{\text{Total no. of papers published in past 2 years.}}$$

Merits of JIF

→ Higher citation Rate

• limitation of h-index -

→ citation is based on total no. of papers published in a journal & not on the basis of quality of papers.

H-Index

h-index = $\frac{\text{Maximum}}{\text{Number of papers}}$

Source Normalized Impact per paper (SNIP)

SNIP is a factor released in 2012 by Elsevier based on scopus to estimate impact of a journal

$$\text{SNIP} = \frac{\text{Raw impact factor (RIP) per paper}}{\text{citation potential / Medium database citation potential}}$$

$$\text{SNIP} = \frac{\text{Journal citation count per article}}{\text{Total no. of articles published in 2012 - 110}}$$

G Index

i10 index: Minimum 10 citations

• Total no. of citations detected in 2012 = 110

$$\text{JIF} = \frac{110}{100} = 1.1$$

• Total no. of citations detected in 2012 = 110

Meta Analysis of Research Findings

It is a quantitative technique used to combine and analyze results from multiple independent studies on the same topic to arrive at a comprehensive conclusion.

Steps involved:

- Define research question and inclusion criteria
- collect and screen relevant studies
- Extract data
- Perform statistical analysis
- Interpret and report findings

Literature

→ A body of work that has been distributed, written or published on a given topic or concept.

→ Various intents: Inform, support, persuade, update or dispute.

→ Consists of

— Major works: key or seminal publication on a topic

continuously evolving topic/writings.

→ Types

1) Primary Lit

2) Secondary Lit — Books, Reports.

3) Tertiary Lit — Book, Wikipedia.

1) Literature Review:

- critical summary of existing research on a particular topic

2) Grouping of Literature

a) Analysis of Literature:

- The objectives and findings
- Methodologies used
- strengths and limitations
- Relevance to your own research

By analyzing you recognize which works are landmark studies, which findings are widely accepted, and where disagreements or gaps exist.

Intellectual Property Rights (IPR):

* It is legal rights that provide creators protection for original works, inventions or the appearance of products, artistic works, scientific developments etc.

* This incentivizes discovery and creativity by providing creators with an opportunity to

Geographical Index
Copyright

Trade marks

IPR

Patents

Designs

Confidential
infor

Module-II

plants

breeder's
rights

circuit
layout

Passing off

Different Types of Referencing

Referencing component

- 1) Authors name
- 2) Title of the paper
- 3) Journal name
- 4) Volume & Issues
- 5) Page Numbers.
- 6) Year of publication.

Different styles → Unit Questions

- 1) APA (American Psychological Association)
- 2) MLA (Modern Language Association)
- 3) Chicago / Turabian
- 4) Harvard
- 5) IEEE
- 6) Vancouver

24/9/2025.

Project Report writing

1. Title Page

2. Declaration and Certificate

3. Acknowledgements: A section where the researcher thanks individuals, organisation or institutions that help during the research process.

4. Abstract: A summary of the research work, including objectives, methods, key findings and significance. snapshot of entire report.

5. Table of Content: A list of all sections and sub sections with page nos.

6. List of figures and Tables: List all figures and tables with their respective page nos.

7. Abbreviations

8. Introduction

9. Literature survey.



10. Time lines and Deliverables

11. Team composition.

- Principal Investigator (PI)

- Co Investigator / collaborator

- External collaborations:

12. Feasibility and Risk Analysis

- Discuss the feasibility of the project in terms of technical, financial and time-related constraints

- Provide a brief risk analysis and mitigation plan for project delays or challenges.

13. References :

- List all books, journal articles, and other academic resources referenced in the proposal
- Follow the appropriate citation format (APA, IEEE, etc)

14. Annexures :

- student / Faculty Bio-data : Attach the detailed CVs of the principal investigator and co-investigator
- Relevant Publications : Attach any previous publications related to the topic.
- Letter of Endorsement : A signed letter from the institution's head or department head, endorsing the Project.

⑧ Technical chapters are must in Thesis writing

Visual Abstract - Go through.

→ 10. Research Methodology :

- Research Design : Description of the design, whether its experimental, simulation based or qualitative.

11. Results.

12. Discussion → interpretation of results

13. Conclusion. Comparison with previous work

Implications

Limitations

Unexpected Results.

13. Conclusion
- List all books, journal articles, and other references referred in the project
 - Summary of findings
 - Contribution to the field

- Future Research

- Practical Application

14. Recommendations

15. References / Bibliography

16. Appendices

- Raw Data

- Supplementary Information

- Code / Algorithms

Formatting Guidelines

- Font : Times New Roman (12 pt)

- Spacing : 1.5 line spacing

- Margins : 1 inch in all sides

- Heading

- Alignment

- Page Numbering

20.09.25

Data visualization

- Plays a crucial role in funded research projects helping to communicate findings, insights and progress to various stakeholders, including funding agencies, collaborators and the research team itself.
- The ability to visually represent data allows for better understanding, interpretation and decision making.

1. Importance of Data Visualization

- * Communication to stakeholders
 - Publications and Dissemination

* Tracking Progress (Project)

Visual tools, such as Gantt chart and performance graphs are frequently used to track milestones and progress against deliverables, which is critical for funded project with predefined timelines and outcomes.

* Analyzing and Interpreting Data

2. Types of data visualisation used in funded Projects:

(1) Quantitative Data Visualisation

→ Line chart : scatterplot

→ Bar chart :

⊗ Ex: convert histogram → bar chart

→ Histogram

* Qualitative Data Visualization

- Word clouds
- Flowcharts and Diagrams
- Concept Maps

3. 3D Visualization

4. Project Progress chart.

- Gantt chart
- Pie chart.

Audience Analysis

It refers to the process of understanding and assessing the characteristics, needs, expectations and interests of the target audience to whom a message or communication is directed.

- Essential for tailoring the presentation of data, results or proposals to meet the specific needs of different stakeholders ensuring, ensuring effective communication and understanding.

Why Audience Analysis is required in Research?

1) Effective Communication:

Research findings can be complex and different stakeholders may have varying levels of expertise.

Audience analysis helps in customizing the presentation of information, ensuring clarity and relevance.

2) Tailored content :

Not all audiences are interested in the same aspects of a research project. Audience analysis allows researchers to focus on the most relevant information for each audience.

2) Improved Persuasiveness

By understanding the expectations and values of the audience, researchers can craft their message.

4) Ethical and cultural considerations.

How audience analysis are conducted in research.

1) Identify the audience.

→ Primary Audience : peer reviewers, supervisors → Direct contact.

→ Secondary Audience : stakeholders influenced by research, funding agencies & public.

→ Tertiary Audience : Broader stakeholders

2) Assess Audience characteristics

→ Demographics : Age, profession, edu level, cultural bg.

→ Expertise and knowledge level

→ Values, interests and priorities.

3.) Analyze Audience Expectations

→ Purpose of the communication

→ Preferred communication channels

→ Desired outcomes

4) Segment the Audience

→ Subgroups

→ Customized Messages

5) Test and Refine Communications

→ Feedback Mechanisms.

Difference in Audience Expectations and Interactions

Communication Styles

Types of Research Publications:

— MNRE

— ISRO.

1) Full-length Research Articles

— Description: comprehensive papers presenting new or original research

— Purpose: To contribute significant new finding in the field

— Length: 8000-10,000 words or more

2) Review Articles

length: 12,000 words or more.

Ref: should consist of 80-100 must.

short communication Report

- concise reports → short length compared to full length papers
- novel discovery in succinct manner
- 1500-3000 words.

4) Case Studies

- These articles provide, a detailed analysis of a specific case or phenomenon often used in medical, legal, business or social science research.

- Real life example in depth.

- length: Usually around 3,000 - 5,000 words.

1) Key characteristics of Research:

Research is a systematic and organized process aimed at generating new knowledge or solving problem. Its key characteristics include:

(i) systematic and structured:

Research follows a methodical approach involving clear steps like defining the problem, collecting data, analyzing and drawing conclusions.

(ii) objective:

It strives for impartiality, relying on evidence and facts rather than personal opinions or biases.

(iii) logical and reproducible

Research is grounded in logical reasoning and its methods should allow others to replicate the study to verify results.

(iv) Empirical:

It is based on observations, experiments or data derived from real-world phenomena.

(iv) Purposeful: selection of a particular problem or question to address. Research aims to address a specific problem, answer a question or contribute to existing knowledge.

(vi) Ethical:

It adheres to ethical standards, ensuring integrity, transparency and respect for participants or subjects.

(vii) Cumulative

Research builds on previous studies, contributing to the existing body of knowledge.

(viii) Critical and Analytical

It involves critical evaluation of data and sources

to ensure accuracy and relevance.

comment on "Literature Review helps in identifying a suitable Research Problem".

The statement is accurate and highlights a critical role of the literature review in the research process. A

literature review involves systematically analyzing existing

studies, theories and findings related to a topic.